

Node 22.17.0 Upgrade — Dependency Compatibility Summary

Date: July 7, 2026

Project: hi-er (HI Energy Ratings)

Target Node version: 22.17.0

Verdict

The current dependency set is **compatible with Node 22.17.0**. No additional dependency changes are required purely for the Node version upgrade.

What Was Checked

Check	Result
<code>engines.node</code> in <code>package.json</code>	Pinned to 22.17.0
Full dependency tree engine constraints	0 mismatches against 22.17.0
Upper-bound Node constraints (< 22, etc.)	None found
<code>npm install --engine-strict</code> on 22.17.0	Succeeds
All 47 direct dependencies <code>require()</code>	All load
<code>@resvg/resvg-js</code> native rendering	Works
App bootstrap (<code>index.js</code>)	Starts and connects to Mongo

Upgrade History

- **409bed5 (Node 22.17.0):** Changed `engines.node` from 19.7.0 to 22.17.0. No dependency version bumps in this commit.
- **cefb728 (Beta 05 26 25):** Upgraded Express from 4.14.0 to 5.1.0, which requires Node ≥ 18. This was the main framework change supporting newer Node versions.

Notable Packages (Old but Working on 22.17.0)

These packages are legacy or unmaintained but still load and run on Node 22.17.0:

Package	Version	Notes
request	2.88.2	Deprecated HTTP client; used in routes/admin.js, routes/participant.js
phantomjs-prebuilt	2.1.14	Abandoned; used with html-pdf for PDF generation in routes/certificates.js
html-pdf	2.2.0	Depends on PhantomJS
connect-mongo	1.3.2	Very old; uses MongoDB driver 2.x while Mongoose 6 uses driver 4.x internally (separate stacks)
express-session	1.14.1	Pre-dates Node 22; no engine blocker
helmet	2.1.3	Pre-dates Node 22; no engine blocker
nodemailer	2.7.0	Pre-dates Node 22; no engine blocker
lodash	4.17.2	Pre-dates Node 22; no engine blocker

Separate Concerns (Not Node Version Blockers)

Security

npm audit reports **94 vulnerabilities** (25 critical, 31 high, 24 moderate, 14 low). These stem from outdated packages, not from Node 22 incompatibility.

Tests

npm test fails with a TypeError in controllers/circulator.js:568 (Cannot read properties of undefined (reading 'push')). This failure occurs on both Node 22.17.0 and 22.18.0 and appears to be an application/test bug, not a Node upgrade issue.

Strict Engine Pin

The engines field is pinned to the exact version "22.17.0". With `npm install --engine-strict`, developers on 22.18.0 or other 22.x patch versions will get an EBADENGINE error. Consider relaxing to "22.x" or "`>=22.17.0 <23`" if patch flexibility is desired.

Recommendations (Future Work)

1. **Security:** Prioritize upgrading or replacing deprecated packages (request, phantomjs-prebuilt/html-pdf, old connect-mongo, etc.).
2. **Tests:** Fix the pre-existing failure in `tests/circulator-model-checks.js` / `controllers/circulator.js`.
3. **Engines field:** Consider relaxing the exact pin if the team wants to allow 22.x patch versions without friction.

Bottom Line

No additional dependency changes are needed purely for Node 22.17.0. Install, require, native modules, and app startup all work. The bigger risks are security debt from unmaintained packages and the pre-existing test failure — not Node version incompatibility.